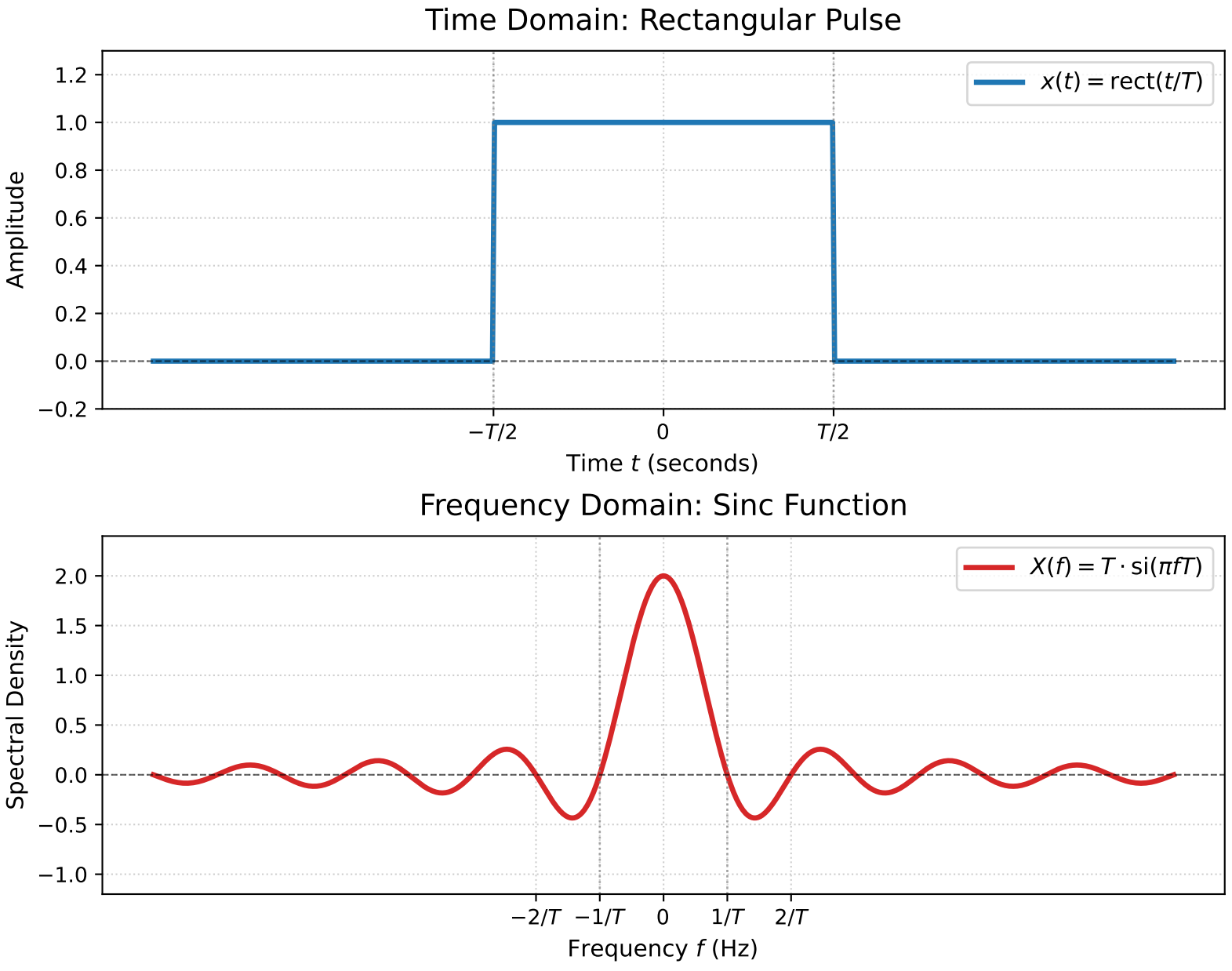
Time Domain: Rectangular Pulse
$x(t) = \text{rect}(t/T)$
Amplitude
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
$-T/2$
0
$T/2$
Time t (seconds)
Frequency Domain: Sinc Function
$X(f) = T \cdot \text{si}(\pi f T)$
Spectral Density
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
$-2/T$
$-1/T$
0
$1/T$
$2/T$
Frequency f (Hz)